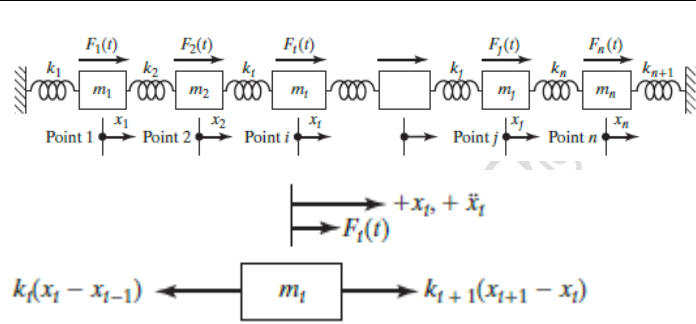


Influence Coefficients

The equations of motion of a multidegree-of-freedom system can also be written in terms of **influence coefficients**. The influence coefficients associated with the stiffness matrix are called **stiffness influence coefficients** and those with mass matrix are known as **inertia influence coefficients**. The inverse of stiffness influence coefficients is known as **flexibility influence coefficients** and the inverse of inertia influence coefficients are known as **inverse inertia coefficients**.

Stiffness influence coefficients

The force required to produce unit elongation in a linear spring is known as stiffness of the spring. The **stiffness influence coefficients** denoted by k_{ij} is defined as the force at point i due to a unit displacement at point j when all the points other than point j are fixed.



The total force F_i at point i due to displacements at all points can be found out by

$$F_i = k_{i1}x_1 + k_{i2}x_2 + \dots + k_{ii}x_i + \dots + k_{in}x_n$$

$k_{i1} \Rightarrow$ is the force at point i due to a unit displacement at point 1 keeping all other displacements zero

$k_{i1}x_1 \Rightarrow$ is the force at point i due to x_1 displacement at point 1

$$F_i = \sum_{j=1}^n k_{ij}x_j \quad : \quad i = 1, 2, \dots, n$$

$$\text{ie., } F_1 = k_{11}x_1 + k_{12}x_2 + \dots + k_{1i}x_i + \dots + k_{1n}x_n$$

$$F_2 = k_{21}x_1 + k_{22}x_2 + \dots + k_{2i}x_i + \dots + k_{2n}x_n$$

$\vdots$

$$F_n = k_{n1}x_1 + k_{n2}x_2 + \dots + k_{ni}x_i + \dots + k_{nn}x_n$$

$$\begin{Bmatrix} F_1 \\ F_2 \\ \vdots \\ F_n \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} & \dots & k_{1n} \\ k_{21} & k_{22} & \dots & k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k_{n1} & k_{n2} & \dots & k_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}$$

Or in the matrix form, it can be written as

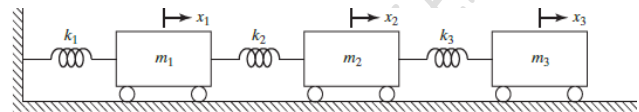
$$\{\vec{F}\} = [k]\{\vec{x}\}$$

$\vec{F}$ is the force vector, $\{\vec{x}\}$ is the displacement vector, and $[k]$ is the stiffness matrix.

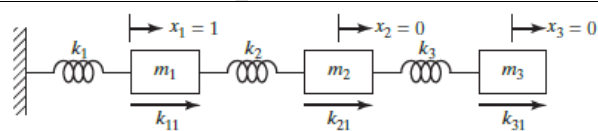
- Since the force required at point i to cause a unit deflection at point j and zero deflections at all other points is same as the force required at point j to produce unit deflection at point i and zero deflections at all other points (This is known as **Maxwell's Reciprocity Theorem**), we can write $k_{ij} = k_{ji}$
- The stiffness influence coefficients k_{ij} of a multi-rotor torsional system can be defined as the torque at rotor i due to a unit angular displacement at rotor j keeping angular displacements at all other rotors equal to zero.
- The stiffness influence coefficients can be calculated by using principles of statics and solid mechanics.

Example 1

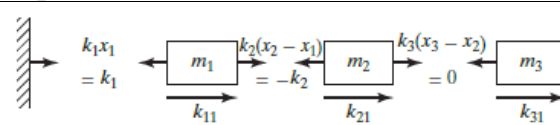
Find the stiffness influence coefficients of the given three degree of freedom system



Assume a force is applied at location 1 to produce a unit deflection at mass 1 and zero displacements at other mass locations. By definition, that force is k_{11}



For making the displacements at locations 2 and 3 equal to zero, some forces must be applied there. The force at location 2 due to a unit displacement at location 1 is defined as k_{21} . Similarly, the force at location 3 due to a unit displacement at location 1 is defined as k_{31} .



From the free body diagrams,

$$\text{Mass 1} \Rightarrow k_{11} + k_2(x_2 - x_1) - k_1x_1 = 0$$

$$\text{Mass 2} \Rightarrow k_{21} + k_3(x_3 - x_2) - k_2(x_2 - x_1) = 0$$

$$\text{Mass 3} \Rightarrow k_{31} - k_3(x_3 - x_2) = 0$$

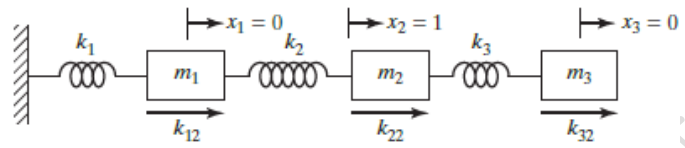
But, $x_1 = 1, x_2 = x_3 = 0$

$$k_{11} = k_1 + k_2$$

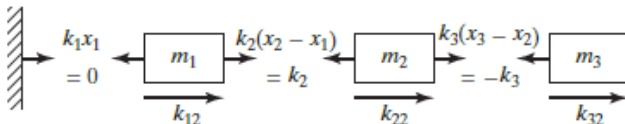
$$k_{21} = -k_2$$

$$k_{31} = 0$$

Next, assume a force is applied at location 2 to produce a unit deflection at mass 2 and zero displacements at other mass locations. By definition, that force is k_{22}



For making the displacements at locations 1 and 3 equal to zero, some forces must be applied there. The force at location 1 due to a unit displacement at location 2 is defined as k_{12} . Similarly, the force at location 3 due to a unit displacement at location 2 is defined as k_{32} .



From the free body diagrams,

$$\text{Mass 1} \Rightarrow k_{12} + k_2(x_2 - x_1) - k_1x_1 = 0$$

$$\text{Mass 2} \Rightarrow k_{22} + k_3(x_3 - x_2) - k_2(x_2 - x_1) = 0$$

$$\text{Mass 3} \Rightarrow k_{32} - k_3(x_3 - x_2) = 0$$

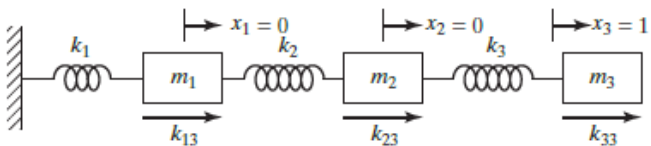
But, $x_2 = 1, x_1 = x_3 = 0$

$$k_{12} = -k_2$$

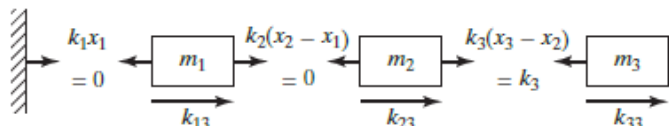
$$k_{22} = k_2 + k_3$$

$$k_{32} = -k_3$$

Again, assume a force is applied at location 3 to produce a unit deflection at mass 3 and zero displacements at other mass locations. By definition, that force is k_{33}



For making the displacements at locations 1 and 2 equal to zero, some forces must be applied there. The force at location 1 due to a unit displacement at location 3 is defined as k_{13} . Similarly, the force at location 2 due to a unit displacement at location 3 is defined as k_{23} .



From the free body diagrams,

$$\text{Mass 1} \Rightarrow k_{13} + k_2(x_2 - x_1) - k_1x_1 = 0$$

$$\text{Mass 2} \Rightarrow k_{23} + k_3(x_3 - x_2) - k_2(x_2 - x_1) = 0$$

$$\text{Mass 3} \Rightarrow k_{33} - k_3(x_3 - x_2) = 0$$

But, $x_3 = 1, x_1 = x_2 = 0$

$$\begin{aligned} k_{13} &= 0 \\ k_{23} &= -k_3 \\ k_{32} &= k_3 \end{aligned}$$

Hence, the stiffness matrix is

$$[k] = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix} = \begin{bmatrix} k_1 + k_2 & -k_2 & 0 \\ -k_2 & k_2 + k_3 & -k_3 \\ 0 & -k_3 & k_3 \end{bmatrix}$$

Flexibility influence coefficients

The flexibility influence coefficients a_{ij} is defined as the deflection at point i due to a unit load at point j . Since we are dealing with linear systems only, the deflection at point i due to a load F_j at point j can be written as $x_{ij} = a_{ij} F_j$

If several forces F_j ($j = 1, 2, \dots, n$) act at different points of the system, the total deflection at any point i can be found by summing up the contributions of all forces F_j

$$\begin{aligned} x_i &= \sum_{j=1}^n x_{ij} = \sum_{j=1}^n a_{ij} F_j, \quad i = 1, 2, \dots, n \\ x_1 &= a_{11}F_1 + a_{12}F_2 + \dots + a_{1i}F_i + \dots + a_{1n}F_n \\ x_2 &= a_{21}F_1 + a_{22}F_2 + \dots + a_{2i}F_i + \dots + a_{2n}F_n \\ &\vdots \\ &\vdots \\ x_n &= a_{n1}F_1 + a_{n2}F_2 + \dots + a_{ni}F_i + \dots + a_{nn}F_n \end{aligned}$$

$$\begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{Bmatrix} F_1 \\ F_2 \\ \vdots \\ F_n \end{Bmatrix}$$

Or in the matrix form, it can be written as

$$\{\vec{x}\} = [a]\{\vec{F}\}$$

$\vec{F}$ is the force vector, $\{\vec{x}\}$ is the displacement vector, and $[a]$ is the flexibility matrix.

$$\{\vec{x}\} = [a]\{\vec{F}\} = [a][k]\{\vec{x}\}$$

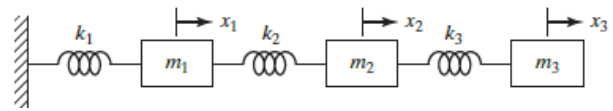
This means that $[a][k] = [I]$

Or $[a] = [k]^{-1}$. Hence, the flexibility matrix is the inverse of stiffness matrix.

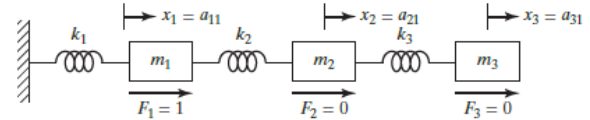
By Maxwell's reciprocity theorem, $a_{ij} = a_{ji}$

Example 2

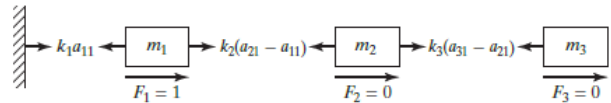
Find the flexibility influence coefficients of the given three degree of freedom system



Assume that a unit force is applied at location 1 and zero forces at other mass locations. By definition, the deflection at mass 1 is $x_1 = a_{11}$



Since the forces at locations 2 and 3 are equal to zero, there will be deflections at locations 2 and 3. The deflection at location 2 due to a unit force at location 1 is defined as $x_2 = a_{21}$. Similarly, the deflection at location 3 due to a unit force at location 1 is defined as $x_3 = a_{31}$.



From the free body diagrams,

$$\text{Mass 1} \Rightarrow F_1 + k_2(x_2 - x_1) - k_1x_1 = 0$$

$$\text{Mass 2} \Rightarrow F_2 + k_3(x_3 - x_2) - k_2(x_2 - x_1) = 0$$

$$\text{Mass 3} \Rightarrow F_3 - k_3(x_3 - x_2) = 0$$

But, $F_1 = 1, F_2 = F_3 = 0$ and $x_1 = a_{11}, x_2 = a_{21}, x_3 = a_{31}$

$$1 = k_1a_{11} - k_2(a_{21} - a_{11}) = (k_1 + k_2)a_{11} - k_2a_{21}$$

$$0 = -k_3(a_{31} - a_{21}) + k_2(a_{21} - a_{11}) = -k_2a_{11} + (k_2 + k_3)a_{21} - k_3a_{31}$$

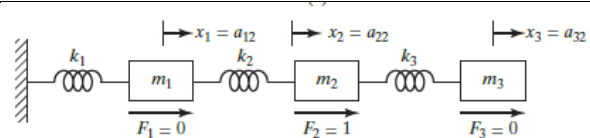
$$k_3(a_{31} - a_{21}) = 0 \Rightarrow a_{31} = a_{21}$$

$$0 = -k_2a_{11} + (k_2 + k_3)a_{21} - k_3a_{21} \Rightarrow k_2a_{11} = k_2a_{21} \Rightarrow a_{11} = a_{21}$$

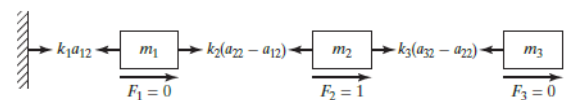
$$1 = (k_1 + k_2)a_{11} - k_2a_{21} \Rightarrow 1 = (k_1 + k_2)a_{11} - k_2a_{11} \Rightarrow a_{11} = \frac{1}{k_1}$$

$$\therefore a_{11} = \frac{1}{k_1} = a_{21} = a_{31}$$

Next, assume a unit force is applied at location 2 and zero forces at other mass locations. By definition, the deflection at mass 2 is $x_2 = a_{22}$



Since the forces at locations 1 and 3 are equal to zero, there will be deflections at locations 1 and 3. The deflection at location 1 due to a unit force at location 2 is defined as $x_1 = a_{12}$. Similarly, the deflection at location 3 due to a unit force at location 2 is defined as $x_3 = a_{32}$.



From the free body diagrams,

$$\text{Mass 1} \Rightarrow F_1 + k_2(x_2 - x_1) - k_1x_1 = 0$$

$$\text{Mass 2} \Rightarrow F_2 + k_3(x_3 - x_2) - k_2(x_2 - x_1) = 0$$

$$\text{Mass 3} \Rightarrow F_3 - k_3(x_3 - x_2) = 0$$

But, $F_2 = 1, F_1 = F_3 = 0$ and $x_1 = a_{12}, x_2 = a_{22}, x_3 = a_{32}$

$$0 = k_1a_{12} - k_2(a_{22} - a_{12}) = (k_1 + k_2)a_{12} - k_2a_{22}$$

$$1 = -k_3(a_{32} - a_{22}) + k_2(a_{22} - a_{12}) = -k_2a_{12} + (k_2 + k_3)a_{22} - k_3a_{31}$$

$$k_3(a_{32} - a_{22}) = 0 \Rightarrow a_{32} = a_{22}$$

$$1 = -k_3(a_{32} - a_{22}) + k_2(a_{22} - a_{12}) \Rightarrow k_2a_{22} - k_2a_{12} = 1$$

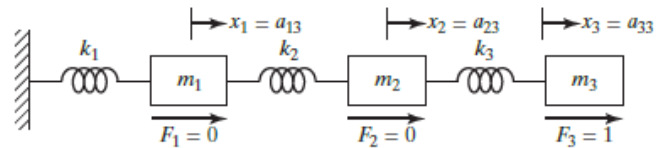
$$0 = (k_1 + k_2)a_{12} - k_2a_{22} \Rightarrow 0 = (k_1a_{12}) + k_2a_{12} - k_2a_{22} \Rightarrow 0 = k_1a_{12} - 1$$

$$\therefore a_{12} = \frac{1}{k_1}$$

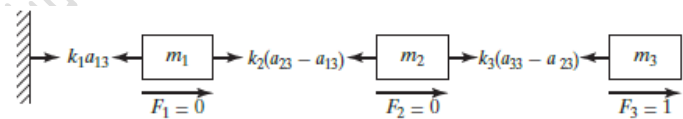
$$k_2a_{22} = 1 + k_2a_{12} = 1 + k_2\left(\frac{1}{k_1}\right) = k_2\left(\frac{1}{k_2} + \frac{1}{k_1}\right) \Rightarrow a_{22} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$a_{32} = a_{22} = \frac{1}{k_1} + \frac{1}{k_2}$$

Again, assume unit force is applied at location 3 and zero forces at other mass locations. By definition, the deflection at mass 3 is $x_3 = a_{33}$



Since the forces at locations 1 and 2 are equal to zero, there will be deflections at locations 1 and 2. The deflection at location 1 due to a unit force at location 3 is defined as $x_1 = a_{13}$. Similarly, the deflection at location 2 due to a unit force at location 3 is defined as $x_2 = a_{23}$.



From the free body diagrams,

$$\text{Mass 1} \Rightarrow F_1 + k_2(x_2 - x_1) - k_1x_1 = 0$$

$$\text{Mass 2} \Rightarrow F_2 + k_3(x_3 - x_2) - k_2(x_2 - x_1) = 0$$

$$\text{Mass 3} \Rightarrow F_3 - k_3(x_3 - x_2) = 0$$

But, $F_3 = 1, F_1 = F_2 = 0$ and $x_1 = a_{13}, x_2 = a_{23}, x_3 = a_{33}$

$$0 = k_1a_{13} - k_2(a_{23} - a_{13}) = (k_1 + k_2)a_{13} - k_2a_{23}$$

$$0 = -k_3(a_{33} - a_{23}) + k_2(a_{23} - a_{13}) = -k_2a_{13} + (k_2 + k_3)a_{23} - k_3a_{33}$$

$$1 - k_3(a_{33} - a_{23}) = 0 \Rightarrow k_3a_{33} - k_3a_{23} = 1$$

$$0 = -k_2a_{13} + k_2a_{23} - 1 \Rightarrow k_2a_{23} - k_2a_{13} = 1$$

$$0 = (k_1 + k_2)a_{13} - k_2a_{23} \Rightarrow 0 = k_1a_{13} + k_2a_{13} - k_2a_{23} \Rightarrow 0 = k_1a_{13} - 1$$

$$\therefore a_{13} = \frac{1}{k_1}$$

$$k_2 a_{23} - k_2 a_{13} = 1 \Rightarrow k_2 a_{23} = k_2 \left(\frac{1}{k_1} \right) + 1 \Rightarrow k_2 a_{23} = k_2 \left(\frac{1}{k_1} + \frac{1}{k_2} \right) \Rightarrow a_{23} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$k_3 a_{33} - k_3 a_{23} = 1 \Rightarrow a_{33} = \frac{1}{k_3} + a_{23} \Rightarrow a_{33} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$$

Hence, the flexibility matrix is

$$[a] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} \left(\frac{1}{k_1} \right) & \left(\frac{1}{k_1} \right) & \left(\frac{1}{k_1} \right) \\ \left(\frac{1}{k_1} \right) & \left(\frac{1}{k_1} + \frac{1}{k_2} \right) & \left(\frac{1}{k_1} + \frac{1}{k_2} \right) \\ \left(\frac{1}{k_1} \right) & \left(\frac{1}{k_1} + \frac{1}{k_2} \right) & \left(\frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} \right) \end{bmatrix}$$

Inertia influence coefficients

The inertia influence coefficients $m_{1j}, m_{2j}, \dots, m_{nj}$ are defined as the set of impulses applied at points 1, 2, ..., n, respectively, to produce a unit velocity at point j and zero velocity at every other point ($\dot{x}_j = 1, \dot{x}_1 = \dot{x}_2 = \dots = \dot{x}_{j-1} = \dot{x}_{j+1} = \dots = \dot{x}_n = 0$). For a multidegree-of-freedom system, the total impulse at point i, P_i , can be found by summing up the impulses causing the velocities $\dot{x}_j$ ($j = 1, 2, \dots, n$) as

$$P_i = \sum_{j=1}^n m_{ij} \dot{x}_j ; \quad i = 1, 2, \dots, n$$

$$P_1 = m_{11}\dot{x}_1 + m_{12}\dot{x}_2 + \dots + m_{1i}\dot{x}_i + \dots + m_{1n}\dot{x}_n$$

$$P_2 = m_{21}\dot{x}_1 + m_{22}\dot{x}_2 + \dots + m_{2i}\dot{x}_i + \dots + m_{2n}\dot{x}_n$$

$$\vdots$$

$$P_n = m_{n1}\dot{x}_1 + m_{n2}\dot{x}_2 + \dots + m_{ni}\dot{x}_i + \dots + m_{nn}\dot{x}_n$$

$$\begin{Bmatrix} P_1 \\ P_2 \\ \vdots \\ P_n \end{Bmatrix} = \begin{bmatrix} m_{11} & m_{12} & \dots & m_{1n} \\ m_{21} & m_{22} & \dots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & \dots & m_{nn} \end{bmatrix} \begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_n \end{Bmatrix}$$

Or in the matrix form, it can be written as

$$\{\vec{P}\} = [m]\{\vec{\dot{x}}\}$$

$\vec{P}$ is the impulse vector, $\{\vec{\dot{x}}\}$ is the velocity vector, and $[m]$ is the inertia influence coefficients matrix.

Inertia influence coefficients matrix is also a symmetric matrix ie; $m_{ij} = m_{ji}$

Potential and Kinetic energy expressions in matrix form

Let x_i denote the displacement of mass m_i and F_i the force applied in the direction of x_i at mass m_i in an n -degree-of-freedom system.

The potential energy or strain energy of the i^{th} spring is given by

$$V_i = \frac{1}{2} F_i x_i$$

Total potential energy,

$$V = \sum_{i=1}^n V_i = \frac{1}{2} \sum_{i=1}^n F_i x_i = \frac{1}{2} \sum_{i=1}^n \left(\sum_{j=1}^n k_{ij} x_j \right) x_i = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n k_{ij} x_j x_i$$

This equation can be written in the matrix form

$$V = \frac{1}{2} \vec{x}^T [k] \vec{x}$$

Kinetic energy associated with mass m_i is given by

$$T_i = \frac{1}{2} m_i \dot{x}_i^2$$

The total kinetic energy,

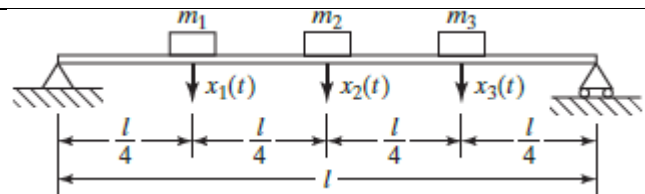
$$T = \sum_{i=1}^n T_i = \frac{1}{2} \sum_{i=1}^n m_i \dot{x}_i^2$$

In the matrix form,

$$T = \frac{1}{2} \dot{\vec{x}}^T [m] \dot{\vec{x}}$$

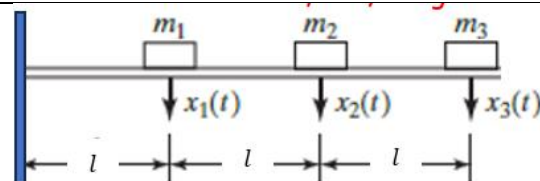
Exercise 1

Determine the stiffness and flexibility influence coefficients of the simply supported beam with negligible mass as shown.



Exercise 1

Determine the stiffness and flexibility influence coefficients of the cantilever beam with negligible mass as shown.



Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition

Vibration & Noise Control..... Dr. Biju N., Professor, School of Engg, CUSAT